WEEK 02

EXPLORATORY DATA ANALYSIS (EDA)

Weekly Internship Progress Report | A. Akshay | Junior Data Scientist Intern

1. Executive Summary

EDA of the Iris dataset (150 rows, four numerical measurements and a species label) was performed using descriptive statistics, species comparisons, histograms, scatter plots, a box plot, and a correlation heatmap. Petal length and width have a strong positive correlation (r = 0.96).

2. Dataset and Source

Measurements are sepal length, sepal width, petal length, and petal width in centimetres for setosa, versicolor, and virginica. Source: UCI Machine Learning Repository — Iris, https://archive.ics.uci.edu/dataset/53/iris

3. Data Overview

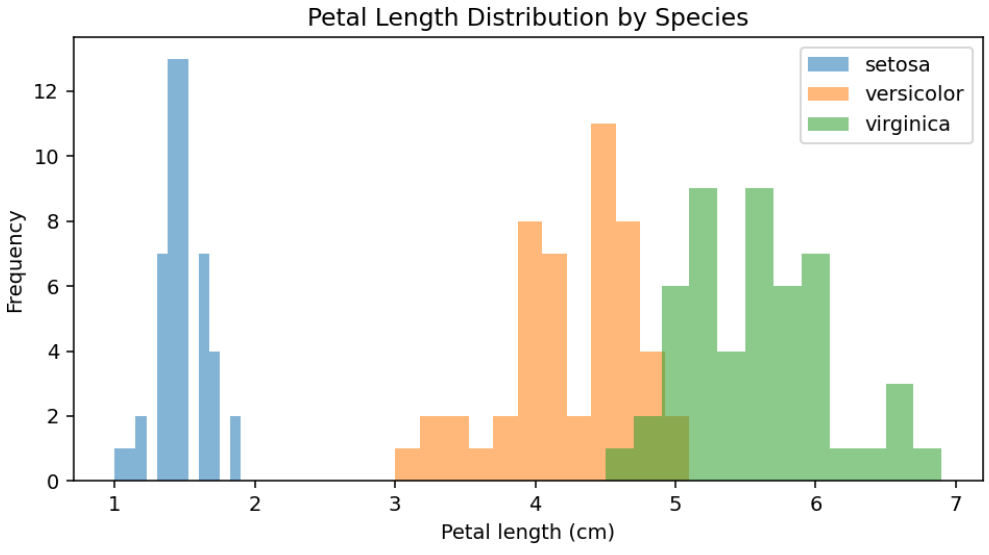
Check	Result
Rows	150
Columns	5
Missing values	0
Duplicate rows	1
Species groups	3

4. Descriptive Statistics

Feature	Mean	Std dev	Min	Max
sepal length cm	5.84	0.83	4.3	7.9
sepal width cm	3.06	0.44	2.0	4.4
petal length cm	3.76	1.77	1.0	6.9
petal width cm	1.20	0.76	0.1	2.5

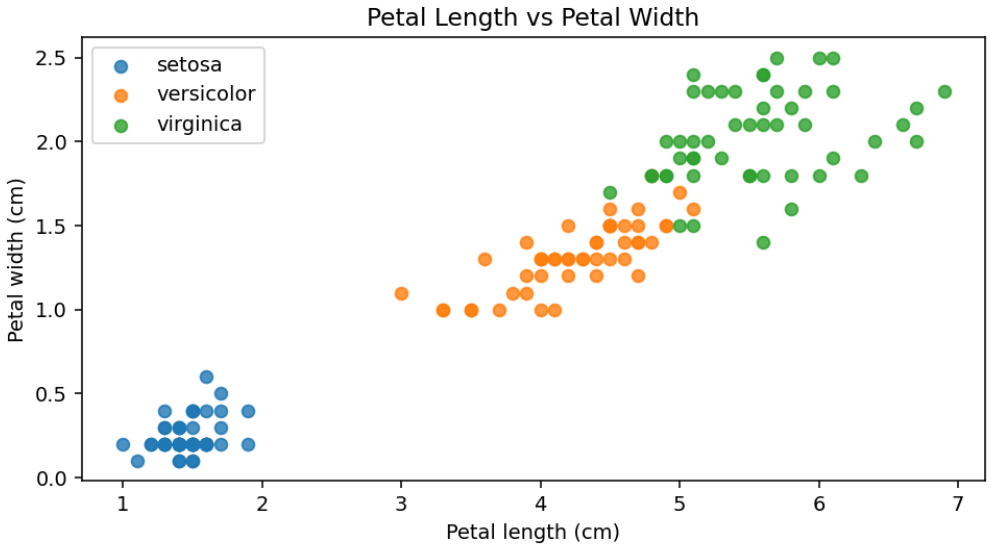
5. Visual Analysis

Petal length histogram



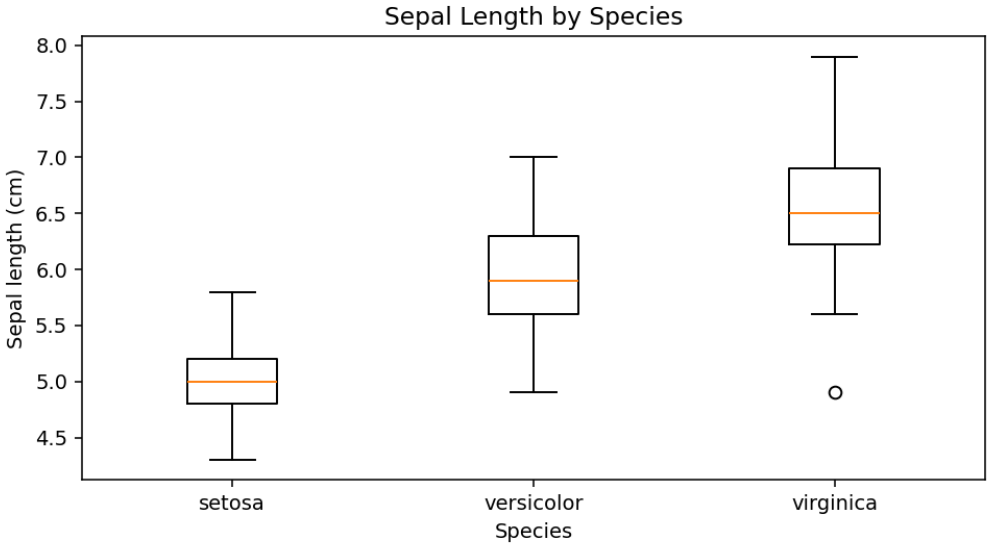
Setosa tends to have shorter petals; the other species tend to have longer petals.

Petal length vs width



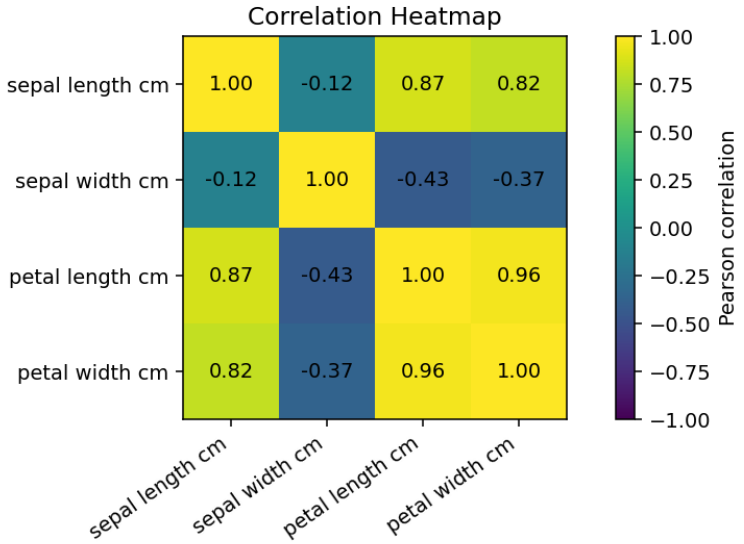
Strong positive association (r = 0.96); visible species clusters.

Sepal length box plot



Species distributions differ but overlap.

Correlation heatmap



Petal length and width have the strongest positive linear relationship; correlation does not prove causation.

6. Species-wise Mean Measurements

Species	Sepal L	Sepal W	Petal L	Petal W
setosa	5.01	3.43	1.46	0.25
versicolor	5.94	2.77	4.26	1.33
virginica	6.59	2.97	5.55	2.03

7. Key Findings, Limitations, and Next Steps

Petal length and width are strongly positively correlated (r = 0.96). Setosa has smaller average petal measurements, while sepal measurements overlap more across species. This is a small educational dataset; findings are descriptive, and no formal significance test was performed. Future work could include hypothesis testing and a baseline classification model.